

# Cardio-oncology electrophysiology

Multichannel modeling of nilotinib-induced ventricular repolarization abnormalities

CORE QUESTION

Does multichannel cardiac ion-channel block explain nilotinib-induced action-potential prolongation differently from hERG-only block, and is the response amplified when repolarization reserve is reduced?

ALEJANDRO

Drug data and pharmacology lead

Data provenance, Hill fitting, uncertainty, quantitative figures.

JUSTIN

Electrophysiology and simulation lead

CiPAORdv1.0/Myokit setup, potassium sensitivity, current scaling, APD and EAD analysis.

12

research weeks

85

published nilotinib measurements

7

cardiac assay currents

Scope: one primary published human ventricular model, one small base-ORd robustness subset, one anticancer drug, one equilibrium concentration-block model, and one prespecified core grid. Dynamic state-dependent hERG fitting is optional future work only.

Workload: 6-8 hours per student per week, plus one 30-45 minute mentor meeting. For every module, the named lead writes the first version and the other student performs documented review.

# What experimental results already exist?

Yes - the project starts with real published experimental measurements. The CiPA dataset contains acute patch-clamp percent-block observations for nilotinib across seven cardiac currents. These are prior experimental data, not results generated by the two students. No CiPAORdv1.0 simulation, APD90, EAD, or hypokalemia result has been generated yet.

| EVIDENCE STATUS                      | AVAILABLE NOW?                  | INTERPRETATION                                               |
|--------------------------------------|---------------------------------|--------------------------------------------------------------|
| Published patch-clamp measurements   | YES - 85 nilotinib observations | Real experimental data from the CiPA channel panel           |
| Student-derived Hill parameters      | NO                              | Generated in Weeks 4 and 10                                  |
| CiPAORdv1.0 action-potential results | NO                              | Generated after model verification in Week 6                 |
| APD90, EAD, and reserve findings     | NO                              | Planned outputs; they must not be presented as known results |

## Published nilotinib concentration-block observations



Concentration (nM, log scale) | colored points: experimental replicates | navy: mean

| Current     | n  | 60 nM          | 300 nM        | 1,800 nM      | 6,000 nM       |
|-------------|----|----------------|---------------|---------------|----------------|
| hERG        | 12 | 40.3 +/- 10.7% | 75.4 +/- 9.1% | 89.3 +/- 5.6% | 98.6 +/- 2.4%  |
| Late sodium | 12 | 1.4 +/- 1.8%   | 3.7 +/- 2.8%  | 13.6 +/- 3.4% | 61.4 +/- 12.9% |
| Calcium     | 12 | 0.7 +/- 0.7%   | 2.0 +/- 0.5%  | 2.0 +/- 0.5%  | 10.5 +/- 5.6%  |
| IKs         | 12 | 1.5 +/- 0.5%   | 2.3 +/- 2.7%  | 1.4 +/- 1.5%  | 9.3 +/- 6.1%   |
| Peak sodium | 12 | 0.0 +/- 0.0%   | 1.0 +/- 1.2%  | 1.2 +/- 2.0%  | 7.2 +/- 5.6%   |
| Kv4.3       | 13 | 0.2 +/- 0.4%   | 0.1 +/- 0.2%  | 1.5 +/- 1.2%  | 7.1 +/- 4.3%   |
| IK1         | 12 | 0.0 +/- 0.0%   | 0.0 +/- 0.0%  | 0.0 +/- 0.0%  | 0.4 +/- 0.8%   |

Table values are descriptive mean +/- sample SD of the raw experimental percent-block observations. Most strata contain three observations; Kv4.3 at 6,000 nM contains four. Crumb et al. reported a free plasma Cmax of 60.4 nM, so the highest measured condition is approximately 100 x free Cmax and must not be described as a typical therapeutic exposure. The lowest measured concentration shows mean hERG block of 40.3%, while the strongest non-hERG signal appears for late sodium at the highest measured concentration. These patterns motivate the planned comparison; they do not themselves establish an action-potential or clinical effect. Source: FDA/CiPA dataset and Crumb et al. (2016).

## Locked scientific design

The cardiac model is ready-made; the students will write only the equilibrium drug-effect and analysis layers. The selected primary model is a frozen endocardial implementation of CiPAORdv1.0, the Dutta-optimized O'Hara-Rudy model. Drug-specific dynamic hERG binding is not fitted or applied in the core project.

| 1. EXPERIMENTAL DATA                                                          | 2. DRUG MODEL                                                                     | 3. CiPAORdv1.0 CELL                                                   | 4. OUTPUTS                                                            |
|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------|-----------------------------------------------------------------------|
| 85 nilotinib patch-clamp observations<br>7 assay currents<br>4 concentrations | Hill concentration-block<br>Primary slope $h = 1$<br>Free slope: sensitivity only | Scale current conductance or permeability<br>Pre-pace to steady state | APD90 primary<br>APD50, triangulation, calcium, EAD/failure secondary |

## Locked input manifests

| DATA MANIFEST                                                                                                                                                                                                                                                                                                                                                                                                       | MODEL MANIFEST                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>File:</b> mergedpatchclampdata-20160514.csv<br><b>Source:</b> FDA/CiPA raw CSV<br><b>Expected:</b> 2,610 total rows; filter Drug == 'Nilotinib' gives 85 rows<br><b>Columns:</b> Drug, Conc, Units, channel, block, pacing<br><b>Rule:</b> block is percent; use <code>block_fraction = block / 100</code> . Preserve raw values and keep the downloaded file read-only. Assay pacing metadata is not model BCL. | <b>Core:</b> CiPAORdv1.0, Dutta-optimized human ventricular endocardial cell<br><b>Source:</b> FDA/CiPA Model-Validation-2018<br><b>Frozen artifact:</b> mentor-validated Myokit .mmt file; record SHA-256 and every variable name in <code>model_manifest.yaml</code><br><b>Engine:</b> Myokit/CVODES; pin Python, Myokit, and dependency versions before analysis<br><b>Boundary:</b> use the optimized cellular model with static equilibrium current scaling; do not fit or apply drug-specific dynamic hERG binding. |

**Assay-to-model map:** hERG -> IKr; Calcium -> ICaL permeability PCa; Peak sodium -> INa; Late sodium -> INaL; IKs -> IKs; IK1 -> IK1; Kv4.3 -> Ito. The mentor must approve exact CiPAORdv1.0 variable names before drug simulation. Base ORd 2011 is reserved for a small Week-11 robustness subset.

## Core equations

$$\text{Block}_j(C) = C^h / (IC50_j^h + C^h)$$

$$\text{Parameter}_j(C) = \text{Parameter}_{j,\text{control}} \times [1 - \text{Block}_j(C)]$$

# Prespecified experiment and expected outcomes

## Prespecified simulation grid

| Factor        | Levels                                       | Interpretation safeguard                                       |
|---------------|----------------------------------------------|----------------------------------------------------------------|
| Concentration | 0; 0.5 x free Cmax; 60, 300, 1,800, 6,000 nM | 0.5 x free Cmax is interpolated; four assay doses are measured |
| Pacing        | Core: BCL 1,000 ms                           | BCL 500 and 2,000 ms move to stretch work                      |
| Drug mode     | hERG-only; multichannel                      | Matched initialization and analysis rules                      |
| Reserve       | K_o 5.4 mM; K_o 3.0 mM if Week-2 gate passes | Hypokalemia is mechanistic sensitivity, not a patient model    |

| EXPECTED SCIENTIFIC PATTERN                                                                                                                                                                                                                                                                                                                                                                                                                                                     | VALID ALTERNATIVE OUTCOMES                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• Nilotinib produces concentration-dependent APD90 prolongation, driven mainly by hERG/IKr block.</li><li>• Near 60 nM, hERG-only and multichannel results may be similar because measured non-hERG block is small.</li><li>• At high laboratory concentrations, late INa block may partly offset hERG-driven prolongation.</li><li>• Hypokalemia may amplify repolarization abnormalities if the Week-2 baseline pilot passes.</li></ul> | <ul style="list-style-type: none"><li>• Multichannel and hERG-only predictions remain similar across the measured range.</li><li>• Competing inward- and outward-current block creates a non-monotonic difference.</li><li>• No EAD occurs; APD90 changes alone remain interpretable.</li><li>• Some high-dose traces show repolarization failure; this is a cellular finding, not clinical arrhythmia evidence.</li></ul> |

**Condition count:** if hypokalemia remains core, 24 nominal combinations collapse to 22 unique conditions after duplicate zero-dose modes are removed. If the pilot fails, the normal-potassium core contains 11 unique conditions rather than 12 nominal combinations.

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NON-NEGOTIABLE | <b>APD90 is primary. Loss of 1:1 capture and failure to repolarize are different events and must be labeled separately: a stimulus that elicits no action potential is capture failure, and repolarization is left un-evaluated; a captured beat that never returns below the APD90 threshold is repolarization failure. Neither is assigned an arbitrary APD. EADs are secondary, and observing no EADs is a valid outcome.</b> |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Weeks 1-2: execution plan

Plan for 6-8 hours per student per week plus one mentor meeting. Each module is written by the named lead, reviewed by the other student, and accepted only after the joint gate is passed.

| WEEK 1      Scientific orientation and reproducible setup                                                                                                                                                                                                                                               |                                                                                                                                                                                                                      |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ALEJANDRO - DATA / PHARMACOLOGY                                                                                                                                                                                                                                                                         | JUSTIN - EP / SIMULATION                                                                                                                                                                                             |
| <ul style="list-style-type: none"> <li>• Create the repository and locked Python environment.</li> <li>• Download/checksum the CiPA CSV and preserve raw values.</li> <li>• Take free Cmax from Crumb et al. Table 1 (newCiPA.csv has no nilotinib row); do not apply protein binding twice.</li> </ul> | <ul style="list-style-type: none"> <li>• Install Myokit and test its compiled solver.</li> <li>• Run the mentor-provided CiPAORdv1.0 model.</li> <li>• Generate one preliminary control action potential.</li> </ul> |
| JOINT GATE                                                                                                                                                                                                                                                                                              | <b>Both students rebuild the environment and explain equilibrium block versus kinetic binding.</b>                                                                                                                   |
| TOOLS / INPUTS                                                                                                                                                                                                                                                                                          | Dutta/ORd and Crumb papers; FDA nilotinib label; Git, Python, pandas, Myokit.                                                                                                                                        |
| REQUIRED READING                                                                                                                                                                                                                                                                                        | Dutta et al. (2017): model optimization; Crumb et al. (2016): Methods and nilotinib table; FDA label: QT and electrolytes.                                                                                           |

  

| WEEK 2      Data audit and model audit                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                |
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| ALEJANDRO - DATA / PHARMACOLOGY                                                                                                                                                                                                         | JUSTIN - EP / SIMULATION                                                                                                                                                                                                                                       |
| <ul style="list-style-type: none"> <li>• Verify 85 nilotinib rows and four concentrations.</li> <li>• Audit channels, units, replicates, missingness, and unusual values.</li> <li>• Create a raw concentration-block panel.</li> </ul> | <ul style="list-style-type: none"> <li>• Locate voltage, calcium, seven currents, K<sub>o</sub>, and scale parameters.</li> <li>• Pilot K<sub>o</sub> 5.4 and 3.0 mM at BCL 1,000 ms.</li> <li>• Freeze the endocardial CiPAORdv1.0 implementation.</li> </ul> |
| JOINT GATE                                                                                                                                                                                                                              | <b>Mentor approves mappings and baseline traces; hypokalemia stays core only if it repolarizes and changes APD90 by <math>\geq 10</math> ms.</b>                                                                                                               |
| TOOLS / INPUTS                                                                                                                                                                                                                          | CiPA CSV; mentor-validated CiPAORdv1.0 .mmt; Myokit; pytest.                                                                                                                                                                                                   |
| REQUIRED READING                                                                                                                                                                                                                        | Dutta et al. (2017): optimized currents; Myokit documentation: models, protocols, simulations, and logging.                                                                                                                                                    |

## Weeks 3-4: execution plan

Plan for 6-8 hours per student per week plus one mentor meeting. Each module is written by the named lead, reviewed by the other student, and accepted only after the joint gate is passed.

| WEEK 3                                                                                                                                                                                                                | Raw pharmacology and control-model verification                                                     |                                                                                                                                                                                                                                                                                                                      |
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| ALEJANDRO - DATA / PHARMACOLOGY                                                                                                                                                                                       |                                                                                                     | JUSTIN - EP / SIMULATION                                                                                                                                                                                                                                                                                             |
| <ul style="list-style-type: none"> <li>Plot every replicate on a log-concentration axis.</li> <li>Show mean and variability without clipping raw values.</li> <li>Flag flat or weakly identified currents.</li> </ul> |                                                                                                     | <ul style="list-style-type: none"> <li>Establish the steady-state procedure at BCL 1,000 ms.</li> <li>Pre-pace <math>\geq 900</math> beats; require APD90 <math>&lt; 0.1</math> ms and peak calcium <math>&lt; 0.1\%</math> over 10 beats.</li> <li>Implement and test APD90 from beat-specific voltages.</li> </ul> |
| JOINT GATE                                                                                                                                                                                                            | Stable control AP, calcium transient, and APD90 before any drug simulation.                         |                                                                                                                                                                                                                                                                                                                      |
| TOOLS / INPUTS                                                                                                                                                                                                        | pandas, Myokit pre-pacing, numpy, pytest, plotting.                                                 |                                                                                                                                                                                                                                                                                                                      |
| REQUIRED READING                                                                                                                                                                                                      | Dutta et al. (2017): control and block validation; Myokit documentation: pacing and state handling. |                                                                                                                                                                                                                                                                                                                      |

| WEEK 4                                                                                                                                                                                                                                 | Hill fitting and current-perturbation tests                                                           |                                                                                                                                                                                                                                           |
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| ALEJANDRO - DATA / PHARMACOLOGY                                                                                                                                                                                                        |                                                                                                       | JUSTIN - EP / SIMULATION                                                                                                                                                                                                                  |
| <ul style="list-style-type: none"> <li>Fit the primary one-parameter Hill model with <math>h=1</math>.</li> <li>Report IC50, residuals, bounds, and fit flags.</li> <li>Use free Hill slope only as a sensitivity analysis.</li> </ul> |                                                                                                       | <ul style="list-style-type: none"> <li>Create a general conductance/permeability scaling function.</li> <li>Unit-test artificial 50% block of each current.</li> <li>Verify that ICaL block scales PCa in this implementation.</li> </ul> |
| JOINT GATE                                                                                                                                                                                                                             | All retained predictions are monotonic and bounded; unreliable fits are explicitly labeled.           |                                                                                                                                                                                                                                           |
| TOOLS / INPUTS                                                                                                                                                                                                                         | scipy.optimize, numpy, pytest, CiPAORdv1.0 parameter map.                                             |                                                                                                                                                                                                                                           |
| REQUIRED READING                                                                                                                                                                                                                       | FDA/CiPA Hill-fitting README; Crumb et al. (2016): concentration-block assay and IC50 interpretation. |                                                                                                                                                                                                                                           |

## Weeks 5-6: execution plan

Plan for 6-8 hours per student per week plus one mentor meeting. Each module is written by the named lead, reviewed by the other student, and accepted only after the joint gate is passed.

| WEEK 5                                                                                                                                                                                                                                 | Drug-model integration and protocol freeze                                                                                                                                                                                                    |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| ALEJANDRO - DATA / PHARMACOLOGY                                                                                                                                                                                                        | JUSTIN - EP / SIMULATION                                                                                                                                                                                                                      |  |
| <ul style="list-style-type: none"> <li>Export a versioned machine-readable parameter file.</li> <li>Implement predicted block for each current and concentration.</li> <li>Verify every predicted block is between 0 and 1.</li> </ul> | <ul style="list-style-type: none"> <li>Implement hERG-only and multichannel modes.</li> <li>Build a batch runner for dose and potassium condition at BCL 1,000 ms.</li> <li>Deduplicate zero-dose modes and implement all outputs.</li> </ul> |  |
| JOINT GATE                                                                                                                                                                                                                             | Freeze 22 unique conditions if the K <sub>o</sub> gate passed, otherwise 11; zero dose is never run twice per potassium condition.                                                                                                            |  |
| TOOLS / INPUTS                                                                                                                                                                                                                         | YAML/CSV parameter file, Myokit, pandas, pytest.                                                                                                                                                                                              |  |
| REQUIRED READING                                                                                                                                                                                                                       | Dutta et al. (2017): pore-block implementation and model optimization; project model/data manifests.                                                                                                                                          |  |

| WEEK 6                                                                                                                                                                                                                                              | Primary concentration-response at normal potassium                                                                                                                                                                                                                                  |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| ALEJANDRO - DATA / PHARMACOLOGY                                                                                                                                                                                                                     | JUSTIN - EP / SIMULATION                                                                                                                                                                                                                                                            |  |
| <ul style="list-style-type: none"> <li>QC the full output table.</li> <li>Plot APD90 versus concentration/free C<sub>max</sub> on a log x-axis.</li> <li>Mark 0.5 x free C<sub>max</sub> as interpolated and measured doses as measured.</li> </ul> | <ul style="list-style-type: none"> <li>Run control, 0.5 x free C<sub>max</sub>, and four measured doses at BCL 1,000 ms.</li> <li>Use normal K<sub>o</sub> 5.4 mM and re-equilibrate every condition.</li> <li>Save final-beat AP, calcium, and selected current traces.</li> </ul> |  |
| JOINT GATE                                                                                                                                                                                                                                          | Both reproduce control, 0.5 x free C <sub>max</sub> , 60 nM, and one high-dose APD90 and identify the interpolated point.                                                                                                                                                           |  |
| TOOLS / INPUTS                                                                                                                                                                                                                                      | Myokit batch runner; frozen Hill parameter file; pandas.                                                                                                                                                                                                                            |  |
| REQUIRED READING                                                                                                                                                                                                                                    | Dutta et al. (2017): simulation protocol and cellular biomarkers; FDA nilotinib label: QT warning.                                                                                                                                                                                  |  |

## Weeks 7-8: execution plan

Plan for 6-8 hours per student per week plus one mentor meeting. Each module is written by the named lead, reviewed by the other student, and accepted only after the joint gate is passed.

| WEEK 7                  | hERG-only versus multichannel comparison                                                                                                                                                                  |                                                                                                                                                                                                                       |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         | <b>ALEJANDRO - DATA / PHARMACOLOGY</b>                                                                                                                                                                    | <b>JUSTIN - EP / SIMULATION</b>                                                                                                                                                                                       |
|                         | <ul style="list-style-type: none"> <li>Calculate change in APD90 from control.</li> <li>Plot multichannel minus hERG-only predictions.</li> <li>Relate differences to measured non-hERG block.</li> </ul> | <ul style="list-style-type: none"> <li>Run matched drug modes with identical initialization rules.</li> <li>Inspect IKr, INaL, and ICaL traces.</li> <li>Audit saved model parameters for every condition.</li> </ul> |
| <b>JOINT GATE</b>       | Each student recreates the main comparison from the other's module; a small difference is acceptable.                                                                                                     |                                                                                                                                                                                                                       |
| <b>TOOLS / INPUTS</b>   | Frozen simulations and parameter audit; numpy and plotting.                                                                                                                                               |                                                                                                                                                                                                                       |
| <b>REQUIRED READING</b> | Li et al. (2019): multichannel versus hERG interpretation and cellular-model limitations.                                                                                                                 |                                                                                                                                                                                                                       |

| WEEK 8                  | Core-grid completion and outcome validation                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                  |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         | <b>ALEJANDRO - DATA / PHARMACOLOGY</b>                                                                                                                                                                                                         | <b>JUSTIN - EP / SIMULATION</b>                                                                                                                                                                                                                                                  |
|                         | <ul style="list-style-type: none"> <li>Audit the BCL 1,000 ms table for duplicates and missing conditions.</li> <li>Blind-review normal, prolonged, and failed traces.</li> <li>Check classifications against saved voltage traces.</li> </ul> | <ul style="list-style-type: none"> <li>Complete the normal-K<sub>o</sub> and hypokalemia grid if the pilot passed.</li> <li>Re-equilibrate each condition and validate APD detection.</li> <li>Separate capture failure from repolarization failure; never flag both.</li> </ul> |
| <b>JOINT GATE</b>       | APD detector passes normal, prolonged, and failure examples; no failed condition is omitted.                                                                                                                                                   |                                                                                                                                                                                                                                                                                  |
| <b>TOOLS / INPUTS</b>   | Deduplicated runner; APD tests; saved traces; manual trace audit.                                                                                                                                                                              |                                                                                                                                                                                                                                                                                  |
| <b>REQUIRED READING</b> | Project APD90 and repolarization-failure definitions; FDA/CiPA simulation-output conventions.                                                                                                                                                  |                                                                                                                                                                                                                                                                                  |

## Weeks 9-10: execution plan

Plan for 6-8 hours per student per week plus one mentor meeting. Each module is written by the named lead, reviewed by the other student, and accepted only after the joint gate is passed.

| WEEK 9                                                                                                                                                                                                                                | Hypokalemia and reduced repolarization reserve                                                              |                                                                                                                                                                                                                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ALEJANDRO - DATA / PHARMACOLOGY</b>                                                                                                                                                                                                |                                                                                                             | <b>JUSTIN - EP / SIMULATION</b>                                                                                                                                                                                 |
| <ul style="list-style-type: none"> <li>Compare <math>K_o</math> 5.4 and 3.0 mM with paired effect plots.</li> <li>Summarize dose-by-potassium patterns.</li> <li>Flag interpretations based only on high laboratory doses.</li> </ul> |                                                                                                             | <ul style="list-style-type: none"> <li>Verify that only extracellular potassium differs.</li> <li>Audit independent steady-state checks under hypokalemia.</li> <li>Save explanatory current traces.</li> </ul> |
| <b>JOINT GATE</b>                                                                                                                                                                                                                     | Complete the core grid and label hypokalemia as mechanistic sensitivity, not a patient-specific prediction. |                                                                                                                                                                                                                 |
| <b>TOOLS / INPUTS</b>                                                                                                                                                                                                                 | Frozen CiPAORdv1.0; $K_o$ 5.4 and 3.0 mM; paired plots.                                                     |                                                                                                                                                                                                                 |
| <b>REQUIRED READING</b>                                                                                                                                                                                                               | FDA label: correct hypokalemia before dosing; Li et al. (2019): repolarization-reserve interpretation.      |                                                                                                                                                                                                                 |

| WEEK 10                                                                                                                                                                                                                      | Experimental uncertainty propagation                                                                |                                                                                                                                                                                                                                        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ALEJANDRO - DATA / PHARMACOLOGY</b>                                                                                                                                                                                       |                                                                                                     | <b>JUSTIN - EP / SIMULATION</b>                                                                                                                                                                                                        |
| <ul style="list-style-type: none"> <li>Bootstrap within every channel-concentration stratum.</li> <li>Create about 200 successful Hill parameter samples.</li> <li>Report fit failures and uncertainty intervals.</li> </ul> |                                                                                                     | <ul style="list-style-type: none"> <li>Propagate samples through the deduplicated BCL 1,000 ms core grid.</li> <li>Record solver failures and runtime.</li> <li>Use lower/median/upper scenarios if runtime is prohibitive.</li> </ul> |
| <b>JOINT GATE</b>                                                                                                                                                                                                            | Explain experimental, parameter, and patient variability; the project estimates only the first two. |                                                                                                                                                                                                                                        |
| <b>TOOLS / INPUTS</b>                                                                                                                                                                                                        | Stratified bootstrap, scipy.optimize, Myokit batch runs.                                            |                                                                                                                                                                                                                                        |
| <b>REQUIRED READING</b>                                                                                                                                                                                                      | FDA/CiPA uncertainty workflow: bootstrap rationale and failure reporting; project fallback rule.    |                                                                                                                                                                                                                                        |

## Weeks 11-12: execution plan

Plan for 6-8 hours per student per week plus one mentor meeting. Each module is written by the named lead, reviewed by the other student, and accepted only after the joint gate is passed.

| WEEK 11          | Base-ORd robustness and mechanistic synthesis                                                                                                                                                                                  |                                                                                                                                                                                                                              |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                  | ALEJANDRO - DATA / PHARMACOLOGY                                                                                                                                                                                                | JUSTIN - EP / SIMULATION                                                                                                                                                                                                     |
|                  | <ul style="list-style-type: none"> <li>Freeze tables and draft Results and Limitations.</li> <li>Visually audit every flagged event.</li> <li>Compare model-variant direction without converting APD to patient QT.</li> </ul> | <ul style="list-style-type: none"> <li>Apply the frozen secondary EAD rule.</li> <li>Run base ORd for control, free-Cmax anchor, and 6,000 nM.</li> <li>Use both drug modes at BCL 1,000 ms and normal potassium.</li> </ul> |
| JOINT GATE       | Report whether the hERG-only versus multichannel conclusion changes between CiPAORdv1.0 and base ORd.                                                                                                                          |                                                                                                                                                                                                                              |
| TOOLS / INPUTS   | Mentor-supplied base ORd; saved traces; audit log; report template.                                                                                                                                                            |                                                                                                                                                                                                                              |
| REQUIRED READING | Li et al. (2019): EAD-related outputs and limitations; FDA label: clinical language versus model language.                                                                                                                     |                                                                                                                                                                                                                              |

| WEEK 12          | Independent reproduction and final presentation                                                                                                                                                                                      |                                                                                                                                                                                                                                             |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                  | ALEJANDRO - DATA / PHARMACOLOGY                                                                                                                                                                                                      | JUSTIN - EP / SIMULATION                                                                                                                                                                                                                    |
|                  | <ul style="list-style-type: none"> <li>Present data provenance, Hill fitting, and uncertainty.</li> <li>Reproduce Justin's simulation pipeline from a clean clone.</li> <li>Answer pharmacology and limitation questions.</li> </ul> | <ul style="list-style-type: none"> <li>Present CiPAORdv1.0 physiology, potassium sensitivity, and EP outcomes.</li> <li>Reproduce Alejandro's data and fitting pipeline.</li> <li>Answer APD, EAD, and model-validity questions.</li> </ul> |
| JOINT GATE       | Both independently run the core workflow and deliver the joint report, presentation, and code package.                                                                                                                               |                                                                                                                                                                                                                                             |
| TOOLS / INPUTS   | Clean clone, locked environment, final report, 10-12 minute presentation.                                                                                                                                                            |                                                                                                                                                                                                                                             |
| REQUIRED READING | Reread Crumb, O'Hara, Dutta, and Li conclusions; verify every final claim against the limitations page.                                                                                                                              |                                                                                                                                                                                                                                             |

# Completion criteria and contingency plan

## Minimum successful project

- A verified CiPAORdv1.0 control simulation and small base-ORd robustness subset.
- A transparent audit of all 85 nilotinib measurements.
- Primary Hill fits with uncertainty and explicit identifiability limitations.
- Deduplicated hERG-only and multichannel simulations at BCL 1,000 ms.
- Normal-potassium versus hypokalemia sensitivity if the Week-2 pilot passes.
- A reproducible report even if multichannel differences are small and no EADs occur.

## Required final deliverables

- Read-only raw-data manifest, processed nilotinib table, and data dictionary.
- Seven replicate-level concentration-block figures and a versioned Hill-parameter table.
- Frozen CiPAORdv1.0 manifest, verified baseline traces, and tested APD90 function.
- APD90-versus-free-Cmax-multiple figure marking interpolated and measured points.
- Hypokalemia sensitivity and base-ORd robustness figures required by their gates.
- Reproducible code repository, final report, and 10-12 minute joint presentation.

## Fallback if an early gate is missed

If the Week 2 model gate or Week 5 integration gate is not passed, freeze the project at one endocardial CiPAORdv1.0 cell, BCL 1,000 ms, normal potassium, control plus the free-Cmax anchor and four measured concentrations, hERG-only versus multichannel block, and APD90 as the only simulation endpoint. Bootstrap uncertainty, EAD detection, hypokalemia, and base-ORd comparison become optional. This reduced project remains a successful undergraduate study.

## Scope limitations

- The patch-clamp data are acute expression-system measurements with small replicate counts.
- The core model represents equilibrium channel block, not dynamic state-dependent hERG binding.
- The project does not model metabolites, chronic channel trafficking, pharmacokinetic variability, tissue propagation, or ECG formation.
- Cellular APD90 is mechanistically related to repolarization but is not the same measurement as a patient's QT interval.
- The  $K_o$  3.0 mM condition is a mechanistic hypokalemia sensitivity analysis, not a model of a specific patient.

# References and reproducibility resources

## Optional future extension

After the core workflow is reproduced, choose at most one: expand the base ORd comparison to the full concentration grid; add BCL 500 and 2,000 ms rate dependence; use pre-fitted dynamic hERG parameters for one oncology drug; or compare one result with ToR-ORd. Do not fit the five kinetic parameters, reconstruct the legacy R/C pipeline, or run 2,000 kinetic bootstrap fits.

## Core references

- [1] O'Hara T, Virag L, Varro A, Rudy Y. Simulation of the undiseased human cardiac ventricular action potential: model formulation and experimental validation. *PLoS Computational Biology*. 2011;7:e1002061. DOI.
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- [3] Crumb WJ Jr, Vicente J, Johannesen L, Strauss DG. An evaluation of 30 clinical drugs against the CiPA proposed ion channel panel. *Journal of Pharmacological and Toxicological Methods*. 2016;81:251-262. DOI.
- [4] FDA/CiPA. Multichannel patch-clamp dataset, mergedpatchclampdata-20160514.csv. [Raw CSV](#).
- [5] FDA/CiPA. Hill fitting workflow and required CSV fields. [README](#).
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- [8] FDA. Nilotinib prescribing information. 2025. [Prescribing information](#).
- [9] Myokit documentation. Models, protocols, simulation, logging, and CellML import. [Documentation](#).
- [10] O'Hara-Rudy 2011 endocardial CellML implementation for the robustness subset. *Physiome Model Repository*. [Model file](#).

Data provenance: downloaded from the FDA/CiPA raw GitHub URL used in Reference 4. Local SHA-256: b5eebc662cf0fcc652cd84674f3b8c77b3b8ac62256a662f6f001107774519b8. Preserve the repository license notice and cite the primary experimental paper when redistributing analyses.